

Week 2 – Assignment 2

Machine Learning: Concepts & Algorithms

Q1. Loan approval prediction — classification or regression?

This is a **classification** problem, since the output is a discrete category (approved or rejected). Among the three algorithms, a **Decision Tree** is most suitable. KNN can also classify, but **Linear Regression** predicts continuous values and is unsuitable for a yes/no decision.

Q2. Feature vs. Label

A **feature** is an input variable used to make a prediction; a **label** is the output we want to predict. *Example:* predicting a house price — the features are area, number of bedrooms, and location, while the label is the price.

Q3. How a Decision Tree reaches a prediction

A Decision Tree splits the data using a series of yes/no questions on features. Starting at the root, it selects the feature that best separates the classes (using a criterion such as Gini impurity or information gain), branches accordingly, and repeats at each node. The customer's feature values follow a path down the branches until reaching a leaf node, whose majority class ("will purchase") becomes the prediction.

Q4. Why is KNN supervised? Can it classify without labels?

KNN is supervised because it requires **labeled training data** — it predicts a new point's label from the known labels of its nearest neighbors. Without labels, KNN **cannot perform classification**, since there are no class labels to vote on. It can only be adapted for unsupervised tasks such as clustering or anomaly detection.

Q5. KNN prediction for $K = 5$

Neighbors: Yes, Yes, No, Yes, No $\rightarrow$ 3 "Yes" and 2 "No". KNN uses **majority voting**, so the prediction is **Yes**, because the most frequent class among the 5 neighbors wins.

Q6. Equation of Linear Regression: $y = mx + b$

- x — input / independent variable (feature)
- y — predicted output / dependent variable
- m — slope (change in y per unit change in x)
- b — intercept (value of y when $x = 0$)

Q7. Correlation

Correlation measures the strength and direction of a linear relationship between two variables.

- +1 — perfect positive correlation (both increase together)
- 0 — no linear relationship
- -1 — perfect negative correlation (one increases as the other decreases)

Q8. Disease present, predicted healthy

The model predicted “Healthy” when the disease was actually present — this is a **False Negative**. It is dangerous because a sick patient is told they are healthy, so they miss treatment, allowing the disease to progress or spread.

Q9. Confusion Matrix vs. Correlation Matrix

A **Confusion Matrix** summarizes a classification model’s performance by counting True Positives, True Negatives, False Positives, and False Negatives, showing where predictions match or differ from actual labels. A **Correlation Matrix** is different: it shows correlation coefficients between pairs of variables (features), measuring relationships within the data itself — not model performance.

Q10. Comparison of algorithms

- **Decision Tree** — splits data into branches using feature-based questions to reach a decision.
- **KNN** — classifies a point based on the majority label of its K closest neighbors.
- **SVM** — finds the optimal hyperplane that maximally separates the classes.
- **Linear Regression** — fits a straight line to model the relationship between inputs and a continuous output.
- **Neural Network** — layers of interconnected neurons that learn complex patterns through weighted connections.